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Robust Modeling for Inference From Generalized Linear Model Classes

Maengseok NOH and Youngjo LEE

Generalized linear models (GLMs) are widely used for data analysis; however, their maximum likelihood estimators can be sensitive to outliers. We propose new statistical models that allow robust inferences from the GLM class of models, including Poisson and binomial GLMs, and their extension to generalized linear mixed models. The likelihood score equations from the new models give estimators with bounded influence, so that the resulting estimators are robust against outliers while maintaining high efficiency in the absence of outliers.

KEY WORDS: Bounded influence; Generalized linear mixed model; Generalized linear model; Robustness.

1. INTRODUCTION

The generalized linear model (GLM) class of models is very attractive for handling a wide variety of continuous and discrete responses. For GLMs, the response variables y_i, for $i = 1, \ldots, n$, follow a distribution belonging to an exponential family such that $E(y_i) = \mu_i$, $\mathrm{var}(y_i) = V(\mu_i)$, and linear predictor

$$\eta_i = g(\mu_i) = \mathbf{x}_i^t \boldsymbol{\beta},$$

where $\boldsymbol{\beta}$ is a p-dimensional vector of parameters and $g(\cdot)$ is a link function. The maximum likelihood estimators (MLEs) for $\boldsymbol{\beta}$ can be obtained by solving the score equations

$$\sum_{i=1}^{n} \frac{y_i - \mu_i}{V(\mu_i)} \frac{\partial \mu_i}{\partial \eta_i} \mathbf{x}_i = 0. \tag{1}$$

But these MLEs have unbounded influence, so that they are sensitive to outliers. Cantoni and Ronchetti (2001) developed robust estimation for GLMs. They considered a general class of M-estimators of Mallows's type, which are the solutions of the amended score equations,

$$\sum_{i=1}^{n} \left\{ \psi_c(r_i) \frac{1}{V^{1/2}(\mu_i)} \frac{\partial \mu_i}{\partial \eta_i} \mathbf{x}_i - \mathbf{a}(\boldsymbol{\beta}) \right\} = 0, \tag{2}$$

where $r_i = (y_i - \mu_i)/V^{1/2}(\mu_i)$, $\psi_c(r_i) = \max(-c, \min(r_i, c))$ is the Huber function, giving an estimator with bounded influence, and

$$\mathbf{a}(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^{n} E(\psi_c(r_i)) \frac{1}{V^{1/2}(\mu_i)} \frac{\partial \mu_i}{\partial \eta_i} \mathbf{x}_i.$$

This approach has been studied for linear mixed models by Huggins (1993), Richardson and Welsh (1995), Richardson (1997), and Gill (2000); here $\mathbf{a}(\boldsymbol{\beta}) = 0$, because $\psi_c(\cdot)$ is an odd function and r_i is symmetric about 0. Extensions to generalized linear mixed models (GLMMs) have been developed by Yau and Kuk (2002) and Sinha (2004). Yau and Kuk (2002) used the penalized quasi-likelihood (PQL), which suffers from severe biases. Because r_i is generally not symmetric in the GLM class of models, the bias-correction term $\mathbf{a}(\boldsymbol{\beta})$ should be computed to ensure the Fisher consistency of the estimator. However, the computation of $\mathbf{a}(\boldsymbol{\beta})$ can be numerically intractable in GLMMs and was done by Sinha (2004) on a model-by-model basis. These methods have information loss when the data are not contaminated.

It is well known that using of the t distribution results in score equations with bounded influence for regression models (Fraser 1979; West 1984; Lange, Little, and Taylor 1989) and linear mixed models (Pinheiro, Liu, and Wu 2001). Mills, Field, and Dupuis (2002) proposed using a mixture of normal and t distributions for random effects. In this article we propose new models, with score equations giving estimators having bounded influence in GLM classes. We also study the data-dependant choice of the tuning parameter. Its advantage is that the resulting robust estimators are consistent, because the MLEs are consistent when the assumed model includes the true model (Sec. 5). To implement likelihood inferences from robust models, we use Laplace approximation based on the h likelihood (Lee and Nelder 1996), which gives an accurate estimate (Huber, Ronchetti, and Victoria-Feser 2004; Lee and Nelder 2006); that is, biases from the approximation could be practically negligible.

We introduce the new models for Poisson and binomial GLMs in Section 2 and their extensions to GLMMs in Section 3. We investigate the choice of tuning parameter in Section 4 and study consistency of the proposed robust estimators in Section 5. In Section 6 we present simulation studies to illustrate the robustness of the proposed methods, and make comparisons with other robust estimators. We apply the proposed methods to the three data sets in Section 7, followed by concluding remarks. Proofs of theorems are collected in the Appendix, and the C++ codes are available at *http://stat.snu.ac.kr/hglmlab/*.

2. BOUNDED–INFLUENCE ESTIMATION FOR GENERALIZED LINEAR MODELS

Consider a hierarchical model $y_i|\sigma_i \sim \mathrm{N}(\mu_i, \sigma_i^2)$, where $\mu_i = E(y_i|\sigma_i) = \mathbf{x}_i^t \boldsymbol{\beta}$. This leads to the linear model

$$y_i = \mathbf{x}_i^t \boldsymbol{\beta} + e_i,$$

where $e_i = \sigma_i z_i$ and $z_i \sim \mathrm{N}(0, 1)$. When $\sigma_i^2 = \sigma^2$, we have the ordinary regression model. Suppose that $\sigma_i^2 = \tau u_i$ with $u_i \sim k/\chi_k^2$ (a kind of the inverse-gamma distribution), and that e_i follows the t distribution, giving robust estimation with bounded

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influence (West 1984; Lange et al. 1989). This approach can be easily extended to gamma GLM classes, which have dispersion parameters σ_i^2. However, in Poisson and binomial GLMs, $\sigma_i^2 = 1$; thus we need another kind of extension.

Suppose that, conditional on unobservable random variables ϕ_i for $i = 1, \ldots, n$, y_i follows a statistical model with

$$\eta_i = g(\mu_i) = \mathbf{x}_i^t \boldsymbol{\beta}, \quad \text{where } \mu_i = E(y_i|\phi_i),$$

such that it becomes the ordinary GLM when $\phi_i = 0$. Let

$$\phi_i = \tau u_i,$$

where u_i has the inverse-gamma distribution with density function

$$f(u_i) = \frac{\tau}{\Gamma(1+1/\tau)} \left(\frac{1}{\tau u_i}\right)^{2+1/\tau} \exp\left(-\frac{1}{\tau u_i}\right), \tag{3}$$

with $E(u_i) = 1$ and $\mathrm{var}(u_i) = \tau/(1-\tau)$ for $\tau \geq 0$.

Here the likelihood of y_i is

$$L_i = \int f(y_i|\phi_i) f(\phi_i)\, d\phi_i.$$

The integration required for this likelihood can be analytically and numerically intractable; thus for inferences about the proposed models, we use Lee and Nelder's (1996, 2006) h-(log-) likelihood given by $h = \sum_{i=1}^n h_i$, where

$$h_i = \log f(y_i|b_i) + \log f(b_i)$$

and $b_i = \log(\phi_i)$. Let $\mathbf{b} = (b_1, \ldots, b_n)^t$. For estimating $\boldsymbol{\beta}$, we use the adjusted profile (log-)likelihood, defined by

$$p_{\mathbf{b}}(h) = \left[h - \frac{1}{2}\log\det\left(\frac{\mathbf{D}(h,\mathbf{b})}{2\pi}\right)\right]\Bigg|_{\mathbf{b}=\tilde{\mathbf{b}}},$$

where $\mathbf{D}(h,\mathbf{b}) = -\partial^2 h/\partial \mathbf{b}^2$ and $\tilde{\mathbf{b}}$ solves $\partial h/\partial \mathbf{b} = 0$. The use of $p_{\mathbf{b}}(h)$ is equivalent to using the Laplace approximation to $l = \sum_{i=1}^n \log L_i$. For robust estimation, we use the solution of the score equation

$$\frac{\partial p_{\mathbf{b}}(h)}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \nu_i(y_i)\mathbf{x}_i = 0, \tag{4}$$

where

$$\nu_i(y_i) = \frac{\partial p_{b_i}(h_i)}{\partial \mu_i}\frac{\partial \mu_i}{\partial \eta_i}.$$

When estimating $\boldsymbol{\beta}$, using h can result in nonignorable biases (Lee and Nelder 1996), which can be avoided by using $p_{\mathbf{b}}(h)$ (Huber et al. 2004; Lee, Nelder, and Pawitan 2006; Noh and Lee 2007).

2.1 Poisson Generalized Linear Models

Suppose that conditional on ϕ_i, y_i follows a negative-binomial distribution with density function

$$f(y_i|\phi_i) = \frac{\Gamma(y_i + 1/\phi_i)}{y_i!\Gamma(1/\phi_i)}\left(\frac{\phi_i\mu_i}{1+\phi_i\mu_i}\right)^{y_i}(1+\phi_i\mu_i)^{-1/\phi_i}. \tag{5}$$

Here $\mathrm{var}(y_i|\phi_i) = \mu_i(1+\phi_i\mu_i)$. When $\tau = 0$, it becomes the Poisson model, with estimating equations (1) for $\boldsymbol{\beta}$ with $V(\mu) = \mu$; this gives unbounded influence as y_i goes to infinity.

Let κ_r^* be the rth-order cumulant of the conditional distribution for $y_i|\phi_i$ and let κ_r be that of the marginal distribution for y_i. The negative-binomial distribution for $y_i|\phi_i$ has

$$\kappa_1^* = \mu_i, \qquad \kappa_2^* = \mu_i(1+\phi_i\mu_i).$$

Here $\phi_i > 0$ represents the overdispersion. When ϕ_i is a fixed constant, higher-order cumulants obey the rule for GLM skewness and kurtosis,

$$\kappa_{r+1}^* = \kappa_2^*(d\kappa_r^*/d\mu_i) \quad \text{for } r = 2, 3.$$

Thus the introduction of fixed ϕ_i does not change the pattern of GLM cumulants κ_{r+1}^* for $r \geq 2$. When ϕ_i is random, the marginal distribution for y_i has mean $\kappa_1 = \mu_i$ and variance $\kappa_2 = E(\kappa_2^*) = \mu_i(1+\tau\mu_i)$, where $\tau > 0$ represents the overdispersion. Furthermore, we have

$$\kappa_3 = E(\kappa_3^*) \geq \kappa_2(d\kappa_2/d\mu_i) \qquad \text{and}$$
$$\kappa_4 = E(\kappa_4^*) + 3\,\mathrm{var}(\kappa_2^*) \geq \kappa_3(d\kappa_3/d\mu_i),$$

because $E(\phi_i^r) \geq \tau^r E(u_i)^r = \tau^r$ for $r \geq 1$. The introduction of the random ϕ_i changes the pattern of higher cumulants; that is, the resulting distribution has larger skewness and kurtosis than the negative-binomial distribution. In this article we are using random effects, ϕ_i, to generate a heavy-tailed Poisson distribution (similar to the heavy-tailed t distribution in regression models); our model is not an ordinary overdispersed Poisson assuming fixed ϕ_i. This is also true for the binomial GLM in Section 2.2. Consequently, the MLEs from the overdispersed negative-binomial distribution have unbounded influence, whereas those from our model do not.

Theorem 1. In Poisson GLM class (5), we have that $\nu_i(y_i) = O(1)$ as y_i goes to infinity.

Figure 1 shows plots of $\nu_i(y_i)$ against y_i for various values of τ at $\mu_i = 5$. Here $\nu_i(y_i)$ converges to a certain value as y_i goes to infinity, so that the estimating equation (4) gives bounded influence if $\tau > 0$.

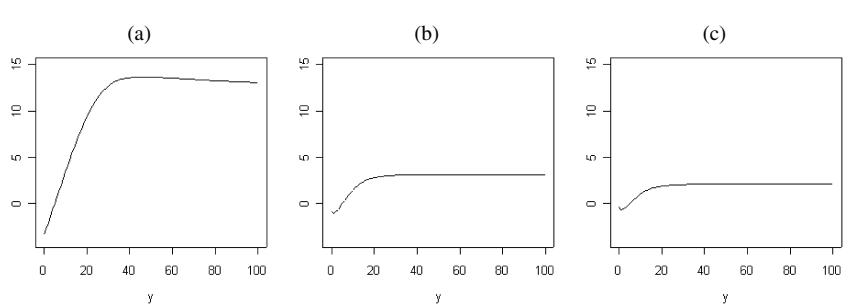

Figure 1. The shape of $\nu_i(y_i)$ for the Poisson model. (a) $\tau = .1$; (b) $\tau = 1$; (c) $\tau = 10$.

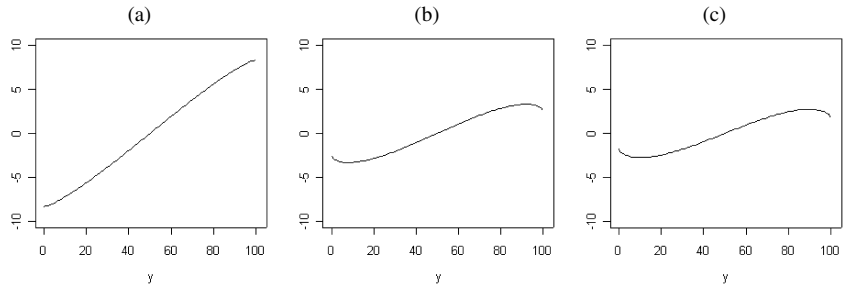

Figure 2. The shape of $v_i(y_i)$ for the binomial model. (a) $\tau = .1$; (b) $\tau = 1$; (c) $\tau = 10$.

2.2 Binomial Generalized Linear Model

Suppose that, conditional on ϕ_i, y_i follows a beta-binomial distribution

$$f(y_i|\phi_i) = \binom{m_i}{y_i} \prod_{k=0}^{y_i-1} (p_i + k\phi_i) \times \prod_{k=0}^{m_i-y_i-1} (1 - p_i + k\phi_i) \Big/ \prod_{k=0}^{m_i-1} (1 + k\phi_i), \quad (6)$$

where $E(y_i|\phi_i) = m_i p_i = \mu_i$ and $\mathrm{var}(y_i|\phi_i) = m_i p_i(1 - p_i) \times \{1 + (m_i - 1)\phi_i/(1 + \phi_i)\}$. When $\tau = 0$, this becomes the binomial model with estimating equations (1) for $\boldsymbol{\beta}$ with $V(\mu) = mp(1 - p)$, which gives negatively (or positively) unbounded influence when y_i approaches 0 (or m_i), with m_i going to infinity.

Theorem 2. In binomial GLM class (6), we have that $v_i(y_i) = O(1)$ when y_i approaches 0 or m_i, with m_i going to infinity.

Figure 2 shows plots of $v_i(y_i)$ against y_i for various values of τ at $m_i = 100$ and $p_i = .5$. Here $v_i(y_i)$ converges to certain values as y_i approaches 0 or m_i with m_i going to infinity, so that estimating equation (4) gives bounded influence when $\tau > 0$.

3. BOUNDED–INFLUENCE ESTIMATION FOR GENERALIZED LINEAR MIXED MODELS

Consider the following GLMM: For $j = 1, \ldots, J_i$ and $i = 1, \ldots, n$,

$$\eta_{ij} = g(\mu_{ij}) = \mathbf{x}_{ij}^t \boldsymbol{\beta} + v_i, \quad (7)$$

where $\mu_{ij} = E(y_{ij}|v_i)$ for responses y_{ij} and $v_i \sim \mathrm{N}(0, \lambda)$. Subscript i designates independent sampling units (subjects), and subscript j designates repeated measurements on each unit. Sinha (2004) extended Cantoni and Ronchetti's (2001) estimating equation (2) for $\boldsymbol{\beta}$ to GLMMs,

$$E\left[\sum_{i=1}^{n}\sum_{j=1}^{J_i}\left\{\psi_c(r_{ij})\frac{1}{V^{1/2}(\mu_{ij})}\frac{\partial\mu_{ij}}{\partial\eta_{ij}}\mathbf{x}_{ij} - \mathbf{a}(\boldsymbol{\beta}, \mathbf{v})\right\}\Big|\mathbf{y}\right] = 0,$$

where $r_{ij} = (y_{ij} - \mu_{ij})/V^{1/2}(\mu_{ij})$, $V(\mu_{ij}) = \mathrm{var}(y_{ij}|v_i)$, and

$$\mathbf{a}(\boldsymbol{\beta}, \mathbf{v}) = \frac{1}{\sum_{i=1}^{n} J_i}\sum_{i=1}^{n}\sum_{j=1}^{J_i} E\{\psi_c(r_{ij})|\mathbf{v}\}\frac{1}{V^{1/2}(\mu_{ij})}\frac{\partial\mu_{ij}}{\partial\eta_{ij}}\mathbf{x}_{ij}.$$

Because of an intractable integration, Sinha (2004) considered models with only one random component. Our method can be easily extended to multicomponent models allowing temporal and spatial correlations for random effects.

We let $\mu_{ij} = E(y_{ij}|v_i, \phi_{ij})$ and

$$\phi_{ij} = \tau u_{ij},$$

where u_{ij} follows the inverse-gamma density with $E(u_{ij}) = 1$ and $\mathrm{var}(u_{ij}) = \tau/(1 - \tau)$ for $\tau \geq 0$. When $\tau = 0$, the resulting model is an ordinary GLMM. Here the h likelihood is given by $h = \sum_{i=1}^{n} h_i$, where

$$h_i = \sum_{j=1}^{J_i}\{\log f(y_{ij}|v_i, b_{ij}) + \log f(b_{ij})\} + \log f(v_i)$$

and $b_{ij} = \log(\phi_{ij})$.

Let $l = \log L$ be a log-likelihood such that $L = \int \exp h \times \prod_{ij} d\phi_{ij} \prod_i dv_i$. For inferences about λ, following Cox and Reid (1987), we may use $p_{\boldsymbol{\beta}}(l)$, which eliminates the nuisance parameter $\boldsymbol{\beta}$ by conditioning on $\tilde{\boldsymbol{\beta}}$. Lee and Nelder (2001) showed that $p_{\mathbf{v},\mathbf{b}}(h)$ is the Laplace approximation to l and furthermore that $p_{\boldsymbol{\beta},\mathbf{v},\mathbf{b}}(h)$ gives a good approximation to $p_{\boldsymbol{\beta}}(l)$, which is the residual or restricted likelihood (Harville 1977) in mixed linear models. For estimation, we use the criteria h for v_i and b_{ij}, $p_{\mathbf{v},\mathbf{b}}(h)$ for $\boldsymbol{\beta}$, and $p_{\boldsymbol{\beta},\mathbf{v},\mathbf{b}}(h)$ for λ. We use the Newton–Raphson method for the maximization. Sinha (2004) showed that these estimating equations for fixed parameters $\boldsymbol{\beta}$ and λ give bounded influence in GLMMs. We show that our models give bounded influence estimation not only for the fixed parameters, but also for the random effects v_i.

3.1 Poisson Generalized Linear Mixed Model

In the Poisson GLMM, the h likelihood estimator $\tilde{v}_i$ for v_i satisfies the relation (see App. C)

$$\tilde{v}_i = O(y_{i+}), \quad \text{with } y_{i+} = \sum_j y_{ij}.$$

Theorem 3. In the Poisson GLMM class, let $\tilde{v}_i$ be the h-likelihood estimator from the robust model. Then

$$\tilde{v}_i = O(1), \qquad \partial p_{\mathbf{v},\mathbf{b}}(h_i)/\partial\beta_k = O(1) \quad \text{for all } k,$$

and

$$\partial p_{\boldsymbol{\beta},\mathbf{v},\mathbf{b}}(h_i)/\partial\lambda = O(1)$$

as y_{i+} goes to infinity.

Thus estimates for both the random effects v_i and the fixed parameters $(\boldsymbol{\beta}^t, \lambda)^t$ have bounded influence against outliers among the y values.

3.2 Binomial Generalized Linear Mixed Model

In the binomial GLMM, the h likelihood estimator $\tilde{v}_i$ for v_i satisfies the relation

$$\tilde{v}_i = O(m_{i+}), \quad \text{with } m_{i+} = \sum_j m_{ij}$$

as y_{i+} approaches 0 or m_{i+}, with m_{i+} going to infinity (see App. C).

Theorem 4. In the binomial GLMM class, let $\tilde{v}_i$ be the h likelihood estimator from the robust model. Then

$$\tilde{v}_i = O(1), \qquad \partial p_{\mathbf{v},\mathbf{b}}(h_i)/\partial \beta_k = O(1) \quad \text{for all } k,$$

and

$$\partial p_{\boldsymbol{\beta},\mathbf{v},\mathbf{b}}(h_i)/\partial \lambda = O(1)$$

when y_{i+} approaches 0 or m_{i+}, with m_{i+} going to infinity.

4. CHOICE OF TUNING PARAMETER τ

In our approach, $1/\tau$ may correspond to the tuning constant c of the Huber function in (2). With the Huber function, as the tuning constant increases, it provides less robust estimators. Thus if τ is large, then it gives more robust estimators against outliers, but with a possible loss of efficiency in the absence of any outliers. Suppose that the response y follows distribution $F(y)$, whereas the estimators $\widehat{\boldsymbol{\beta}}$ are obtained by maximizing the assumed log-likelihood l. Let b be the bound on the influence function of the estimators $\widehat{\boldsymbol{\beta}}$,

$$b = \sup_y \|\mathbf{IF}\|,$$

where $\mathbf{IF} = -\{\int(\partial^2 l/\partial\beta\,\partial\boldsymbol{\beta}^t)\,dF(y)\}^{-1}(\partial l/\partial\boldsymbol{\beta})$. Cantoni and Ronchetti (2001) and Sinha (2004) used the value $c = 1.2$ in (2), satisfying $b \le 10.45$ in the damaged carrots data of Section 7.2; in our approach, this corresponds to $\tau = .37$. To compare our method with those of Cantoni and Ronchetti (2001) and Sinha (2004), we use the value $\tau = .37$. Yau and Kuk (2002) used the value $c = 2.0$.

In this article we may estimate τ by using the restricted ML (REML) estimator that maximizes $p_{\boldsymbol{\beta},\mathbf{b}}(h)$ for GLMs and $p_{\boldsymbol{\beta},\mathbf{b},\mathbf{v}}(h)$ for GLMMs. A motivation for this is as follows. When τ is estimated close to 0, the ordinary GLM class ($\tau = 0$) is a good representation for the data. Thus, in the absence of an outlier, the REML estimator for τ tends to 0, maintaining high efficiency, while in the presence of large outliers, it tends to be larger, giving robust estimations. Using a simulated data set from the Poisson and binomial GLMs in Section 6.1, we set outliers $y_i + k$ for $k \ge 0$. Figure 3 shows changes of $\|\mathbf{IF}\|$ against values of k. The resulting influence function of estimator converge as k increases, showing that using REML estimation for τ provides an estimator with bounded influence, as does using a fixed value.

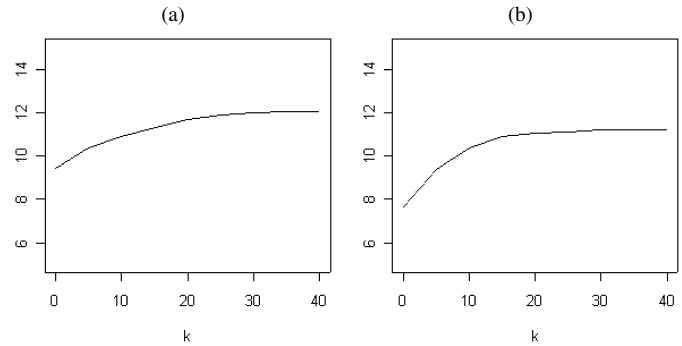

Figure 3. $\|\mathbf{IF}\|$ of the robust estimator $\hat{\boldsymbol{\beta}}$ when the tuning parameter is estimated by the REML method in Poisson (a) and binomial (b) GLMs.

5. CONSISTENCY OF ROBUST ESTIMATORS

Let $l(\boldsymbol{\theta}, \tau) = \sum_{i=1}^n \log f(y_i; \boldsymbol{\theta}, \tau)$ be the log-likelihood, where $\boldsymbol{\theta}$ are parameters and τ is a tuning parameter. In this section we study the Fisher consistency of proposed robust estimators when $\tau = 0$. Proofs and the necessary regularity conditions are given in Appendix D.

Theorem 5. In Poisson GLM class (5) of Section 2.1 and Poisson GLMM class of Section 3.1, the robust estimators for $\boldsymbol{\theta}$ with a fixed tuning parameter $\tau = \tau_0$, maximizing the likelihood $l(\boldsymbol{\theta}, \tau_0)$, are consistent for all of the models indexed by $\tau \ge 0$ as n goes to infinity.

Robust estimators for $\boldsymbol{\beta}$ from t distributions with a fixed tuning parameter value $\tau_0 > 0$ are consistent (as n goes to infinity) in normal distribution (when $\tau = 0$). This result shows that in the Poisson GLM class, the robust estimators with fixed tuning parameter value $\tau_0 > 0$ are also consistent when $\tau = 0$; however, this is not generally the case in binomial GLM class (6) of Section 2.2.

Theorem 6. In binomial GLM class (6) of Section 2.2 and the binomial GLMM class, the robust estimators for $\boldsymbol{\theta}$ with a fixed tuning parameter $\tau = \tau_0$, maximizing the likelihood $l(\boldsymbol{\theta}, \tau_0)$, are consistent for all of the models indexed by $\tau \ge 0$ if in the binomial GLM class both $\min\{m_i\}$ and n go to infinity and $0 < p_i < 1$ for all i, and in the binomial GLMM class both $\min\{m_{ij}\}$ and n go to infinity and $0 < p_{ij} < 1$ for all i and j.

Theorem 7. The REML estimator $\hat{\tau}$ maximizing $l(\boldsymbol{\theta}, \tau)$ converges to zero in probability when $\tau = 0$ as n goes to infinity.

This theorem shows that using REML estimator for τ makes the resulting robust estimators always consistent when $\tau = 0$. Thus, in the binary GLM class with $m_i = 1$, the robust estimators with fixed $\tau = \tau_0 > 0$ are not consistent when $\tau = 0$, whereas those with the data-adaptive choice of the tuning parameter are consistent. This illustrates an advantage of the data-adaptive choice of the tuning parameter.

6. SIMULATION STUDIES

6.1 Generalized Linear Models

We consider the two cases (contaminated and noncontaminated) for Poisson and binomial GLMs. We simulate the Poisson GLM with sample sizes $n = 30, 100$ and linear predictor

$$\eta_i = \log(\mu_i) = \beta_0 + \beta_1 x_i,$$

where $\beta_0 = \beta_1 = 1$. We simulate the binomial GLM with $n = 30, 100$, a binomial denominator $m = 40$, and

$$\eta_i = \text{logit}(p_i) = \beta_0 + \beta_1 x_i,$$

where $\beta_0 = -1$ and $\beta_1 = 1$. To generate the contaminated samples, we randomly choose 5% of the data chosen to make outliers. We replace the corresponding response y by $y + 10$ in the Poisson GLM and by $\min(y + 10, m)$ in the binomial GLM. Such outliers are referred to as mean-shift outliers, and classical MLEs are often vulnerable to them (Cook and Weisberg 1982). The covariate x_i is generated from an independent standard normal distribution. For comparison of the various methods, we compute the finite-sample relative efficiency (RE), defined by

$$RE = \frac{\text{MSE of MLE}}{\text{MSE of robust estimator}},$$

where MSE denotes mean squared error. We include the robust estimator from Cantoni and Ronchetti (2001) for comparison with the new method. Based on 200 replications of simulated data, the simulation results for β_1 are summarized in Figure 4. The cases in which the bias is significant at 5% are designated by *. Note that the MLEs from the GLMs have the largest bias, resulting in the lowest efficiency in the contaminated data. Cantoni and Ronchetti's estimators perform similar to our robust estimator with fixed τ. We prefer to use the adaptive REML estimator to derive the tuning parameter because it gives robust estimation for contaminated data without losing efficiency for noncontaminated data. The same conclusion is reached for β_0 (results not reported).

Both Cantoni and Ronchetti's (2001) and Sinha's (2004) methods need to compute the bias-correction term, whereas our method needs to estimate extra parameters $(\tau, \phi_1, \ldots, \phi_n)$. It is of interest to compare the speed of these methods. We implemented our methods and Yau and Kuk's (2002) method using the C++ programming language and Cantoni and Ronchetti's method using their R code, available at *http://www.unige.ch/ses/metri/cantoni/publications.html*. For Sinha's method, we use the simulation results from his article for comparison. On a Pentium IV 3.2-GHz PC with 512 MB RAM, fitting all of the simulated data in this section took 2.4 hours by robust methods with fixed τ, 2.6 hours by Cantoni and Ronchetti's method, and 6.8 hours by the robust method using REML for τ: Because Cantoni and Ronchetti's method uses the fixed tuning constant, it corresponds to our robust method with fixed τ. Even though our robust method with fixed τ is slightly faster than Cantoni and Ronchetti's method, it may not be meaningful because they use different languages. Regardless, we may say that they are comparable. REML estimation for τ makes the robust procedure more than twice as slow. For GLMMs, Sinha (2004) used the Monte Carlo Newton–Raphson method, which is much slower than Cantoni and Ronchetti's method for GLMs. The run time of our method does not increase much in GLMMs.

6.2 Generalized Linear Mixed Models

We investigated three different scenarios. We first considered the mean-shift outliers mentioned in the previous section. Con-

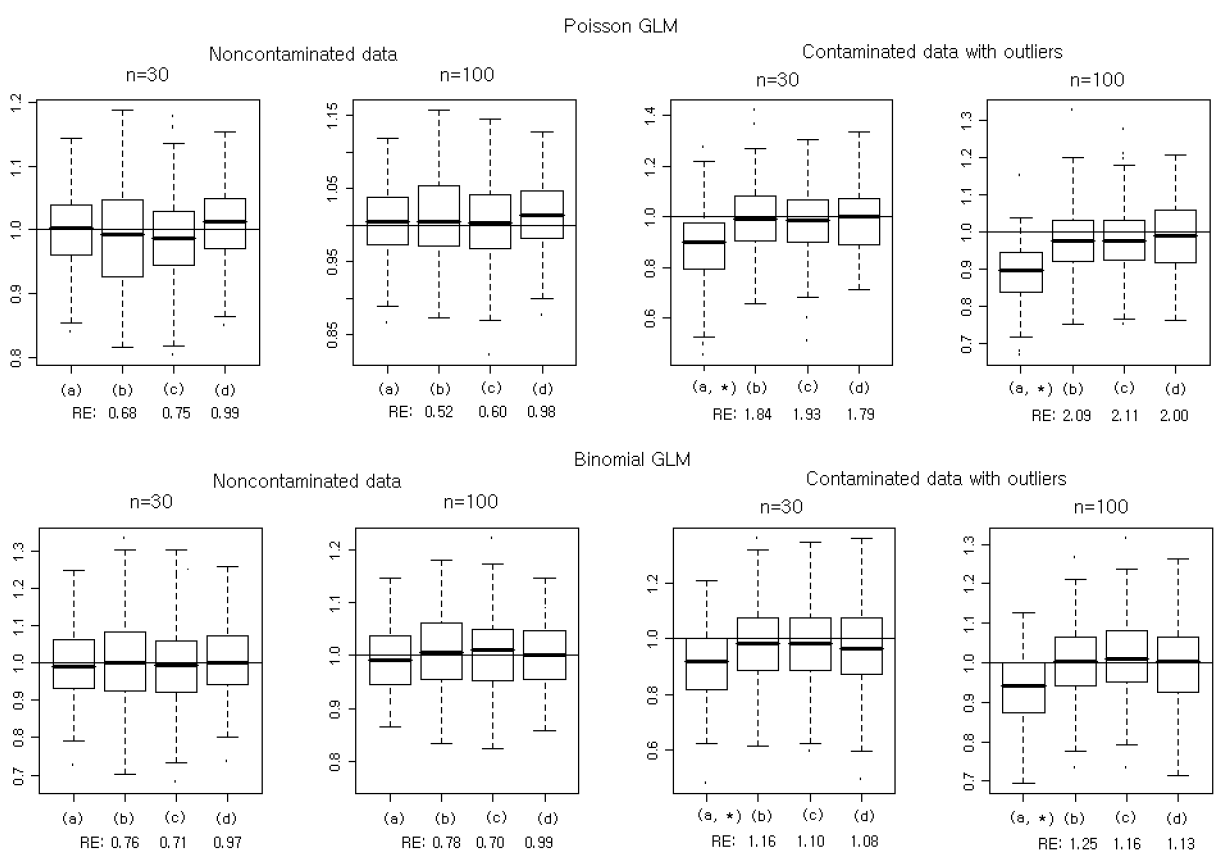

Figure 4. Simulation results for β_1 in the GLM. Boxplots for (a) the classical ML method, (b) Cantoni and Ronchetti's method, (c) the robust method using fixed τ, and (d) the robust method using REML are plotted. *Cases in which the observed bias is significant at the 5% level.

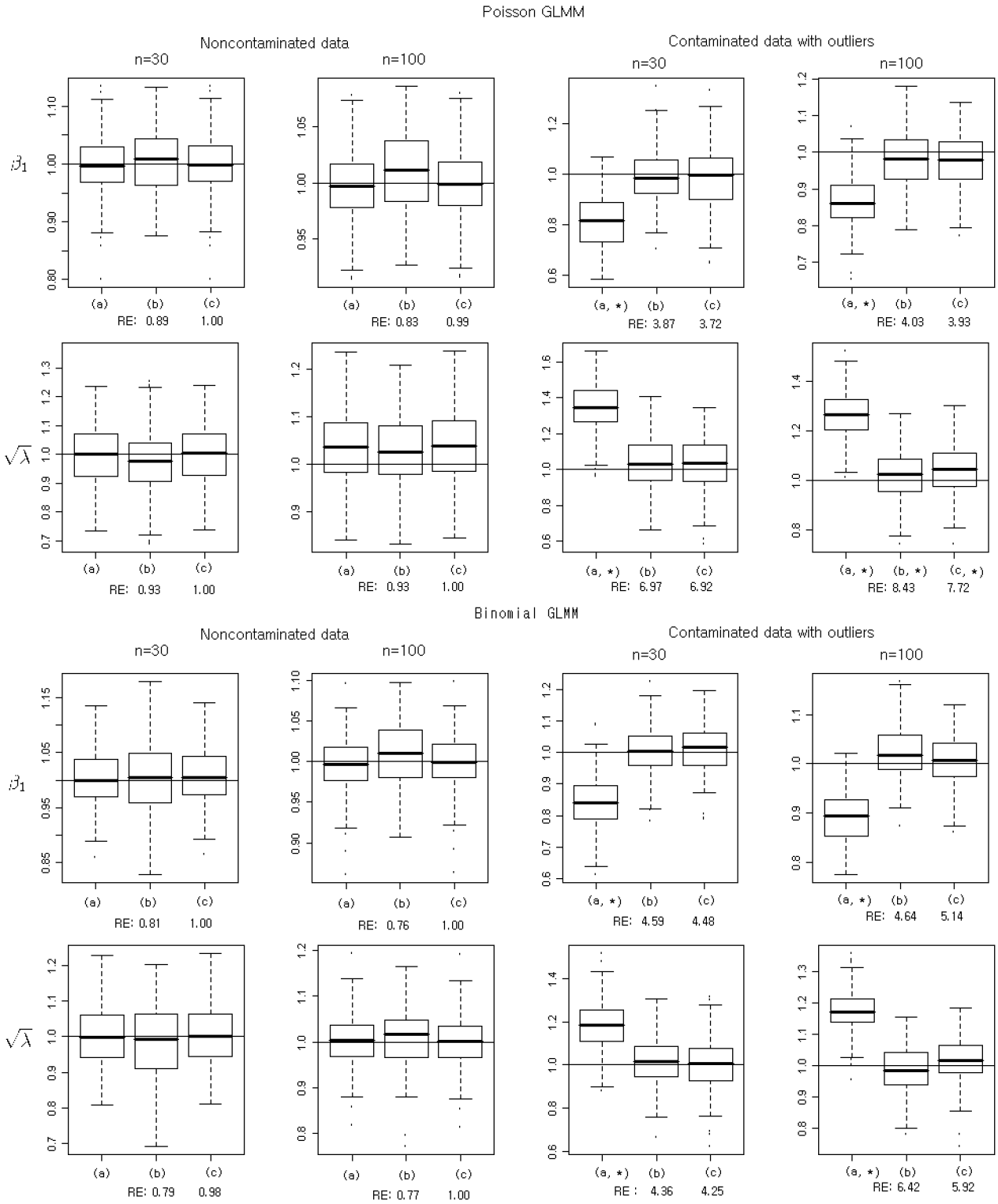

Figure 5. Simulation results for β_1 and $\sqrt{\lambda}$ in the GLMM. Box plots for (a) the classical ML method, (b) the robust method using fixed τ, and (c) the robust method using REML. *Cases in which the observed bias is significant at the 5% level.

sider GLMMs with $n = 30, 100$, $J_i = 3$, and linear predictor

$$\eta_{ij} = \beta_0 + \beta_1 x_{ij} + v_i, \tag{8}$$

where $v_i \sim \mathrm{N}(0, \lambda)$. From Figure 5, we see that the MLE from GLMMs works well in noncontaminated cases but has the largest bias in contaminated data. The proposed estimators are robust against the mean-shift outliers. Using the REML estimator for the tuning parameter gives robust estimation, maintaining high efficiency for noncontaminated data.

We investigated the sensitivity of robust estimators compared with nonrobust estimators as follows. Using one data set from the foregoing simulation for the Poisson GLMM, we set outliers $y_{1j} + k$ for $j = 1, 2, 3$. Figure 6 shows changes of parameter

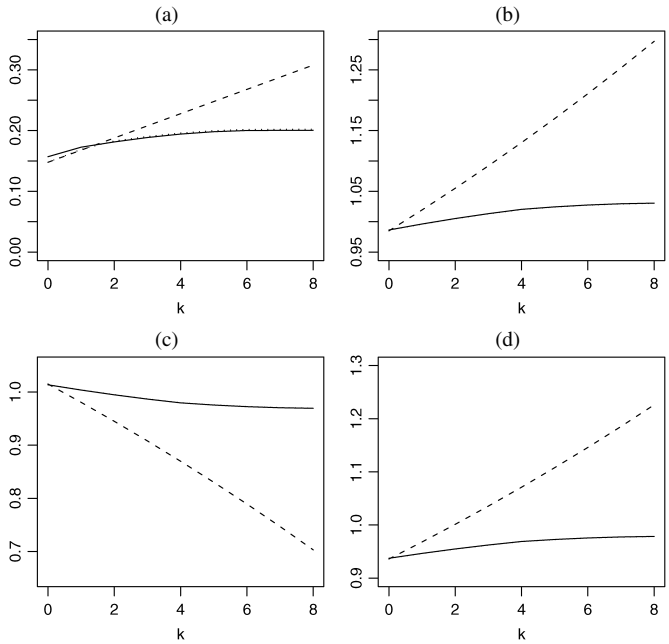

Figure 6. Sensitivity curves of the $\tilde{v_1}$ (a), $\widehat{\beta_0}$ (b), $\widehat{\beta_1}$ (c), and $\widehat{\lambda}$ (d) in the Poisson GLMM for the classical method (- - - - - -), the robust method using fixed τ (——), and the robust method using REML (· · · · · ·).

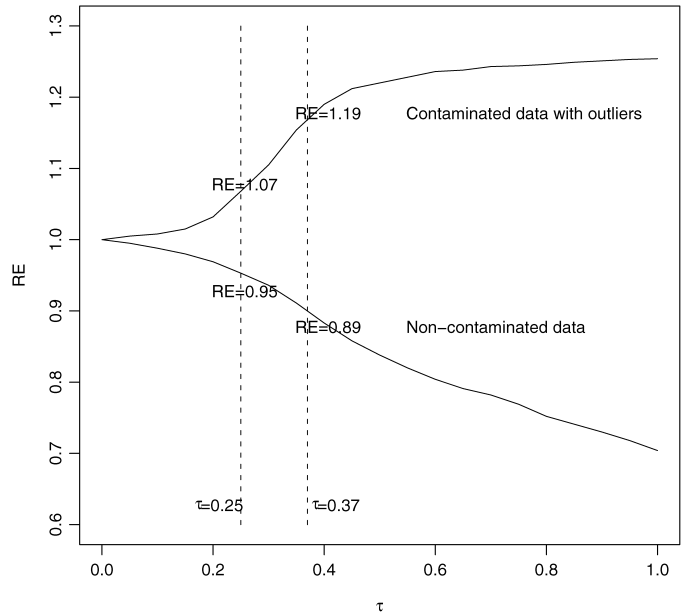

Figure 7. The averages of RE with respect to τ.

and random-effect estimates against values of k. The estimates from the nonrobust method continue to increase or decrease as k increases, whereas the robust estimators converge. The two robust estimators are not distinguishable in the graphs.

Yau and Kuk (2002) studied contamination of random effects. We consider a Poisson GLMM (8) with a log link, $n = 50$, and $J_i = 2$. For simulation x_{ij} was chosen randomly as 0 or 1, $v_i \sim \mathrm{N}(0, \lambda)$, $\beta_0 = -.5$, $\beta_1 = 1$, and $\lambda = 1$. Yau and Kuk considered a mixture distribution for the random effects by taking $v_i \sim (1-\delta)\mathrm{N}(0, \lambda) + \delta\mathrm{N}(0, 11)$. They took $\delta = 0$ for the noncontaminated data and $\delta = .1$ for the contaminated data. Based on 200 replications of simulated data, they computed the mean, standard deviation (SD) and RE for robust and nonrobust (ML) estimators. The results, given in Table 1, show that their estimators are severely biased for small J_i and $1/\lambda$. Even in contaminated cases, their estimator is less efficient than the classical ML method due to large biases. Table 1 shows that our robust methods do not suffer such biases. We again prefer to use the REML estimator because it maintains high efficiency in noncontaminated data. Their method is slightly faster than our robust method with fixed $\tau = .37$ (see Sec. 6.3).

An anonymous referee pointed out that the value of τ must be chosen data-dependently, and that most often it is chosen to reach a certain level of efficiency such as 95%. We compute its average RE for $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\lambda}$ using simulations with respect to τ (Fig. 7). In this example, $\tau = .25$ maintains 95% of the RE in noncontaminated data. Smaller values of τ give a higher RE in noncontaminated data, but a lower RE in contaminated data. When $\tau = .25$ (.37), the average RE is .95 (.89) in noncontaminated data and 1.07 (1.19) in contaminated data. Using the adaptive REML estimator gives an average RE of 1.20 in contaminated data while maintaining high efficiency, .98, in noncontaminated data. In this article we use $\tau = .37$ for the rest of the data sets to compare our method with the methods of Cantoni and Ronchetti (2001) and Sinha (2004).

Table 1. Simulation results for Yau and Kuk's study with $\beta_0 = -.5$ and $\beta_1 = \lambda = 1$

	Classical ML method	Yau and Kuk's method		Robust method using fixed $\tau = .37$		Robust method using REML		Robust method using fixed $\tau = .25$	
	Mean (SD)	Mean (SD)	RE	Mean (SD)	RE	Mean (SD)	RE	Mean (SD)	RE
Noncontaminated data ($\delta = 0$)									
β_0	$-.51_{(.32)}$	$-.38^*_{(.34)}$	.79	$-.52_{(.35)}$	.83	$-.50_{(.32)}$	.98	$-.51_{(.33)}$	.97
β_1	$1.01_{(.43)}$	$.88^*_{(.45)}$	.85	$1.01_{(.45)}$	.91	$1.01_{(.43)}$	.98	$1.01_{(.44)}$	.96
λ	$1.02_{(.33)}$	$.85^*_{(.34)}$	.79	$.99_{(.34)}$	.94	$1.01_{(.34)}$	.97	$1.01_{(.33)}$	.93
Contaminated data with outliers ($\delta = .1$)									
β_0	$-.41^*_{(.31)}$	$-.36^*_{(.31)}$	.90	$-.50_{(.31)}$	1.08	$-.50_{(.31)}$	1.09	$-.48_{(.31)}$	1.03
β_1	$.93^*_{(.42)}$	$.80^*_{(.42)}$	.84	$.99_{(.42)}$	1.03	$.99_{(.42)}$	1.03	$.98_{(.42)}$	1.01
λ	$.79^*_{(.30)}$	$.74^*_{(.29)}$	.88	$.97_{(.30)}$	1.48	$.98_{(.30)}$	1.49	$.90^*_{(.30)}$	1.18

*Cases in which the observed bias is significant at the 5% level.

Table 2. Simulation results for Sinha's (2004) study

Parameter	Classical ML method Mean (SD)	Sinha's method Mean (SD)	Sinha's method RE	Robust method using fixed τ Mean (SD)	Robust method using fixed τ RE	Robust method using REML Mean (SD)	Robust method using REML RE	Robust method using fixed τ and w Mean (SD)	Robust method using fixed τ and w RE	Robust method using REML and w Mean (SD)	Robust method using REML and w RE
Poisson GLMM											
Noncontaminated data											
$\beta_1 = 1$	$1.00_{(.10)}$	$1.00_{(.10)}$	1.00	$1.00_{(.10)}$	1.00	$1.00_{(.10)}$	1.00	$1.00_{(.10)}$	1.00	$1.00_{(.10)}$	1.00
$\lambda = .25$	$.25_{(.07)}$	$.25_{(.07)}$	1.00	$.25_{(.07)}$	1.00	$.25_{(.07)}$	1.00	$.25_{(.07)}$	1.00	$.25_{(.07)}$	1.00
Contaminated data											
$\beta_1 = 1$	$.88^*_{(.11)}$	$.98_{(.10)}$	2.55	$.91^*_{(.10)}$	1.46	$.94^*_{(.11)}$	1.69	$.97_{(.11)}$	2.30	$.98_{(.11)}$	2.12
$\lambda = .25$	$.26_{(.06)}$	$.26_{(.06)}$	1.02	$.26_{(.06)}$	1.02	$.26_{(.06)}$	1.01	$.26_{(.06)}$	1.01	$.26_{(.06)}$	1.01
Binary GLMM											
Noncontaminated data											
$\beta_1 = 1.5$	$1.51_{(.19)}$	$1.52_{(.19)}$	.94	$1.53_{(.21)}$	.87	$1.51_{(.19)}$	1.00	$1.53_{(.21)}$	.86	$1.51_{(.19)}$	1.00
$\lambda = 1$	$1.03_{(.45)}$	$1.06_{(.49)}$	.83	$1.06_{(.48)}$	.85	$1.03_{(.45)}$	1.00	$1.06_{(.48)}$	.84	$1.03_{(.45)}$	1.00
Contaminated data											
$\beta_1 = 1.5$	$1.18^*_{(.14)}$	$1.47_{(.21)}$	2.71	$1.21^*_{(.18)}$	1.17	$1.28^*_{(.18)}$	1.51	$1.47_{(.22)}$	2.48	$1.48_{(.22)}$	2.50
$\lambda = 1$	$.82^*_{(.41)}$	$1.17^*_{(.55)}$	.61	$1.14^*_{(.53)}$	.64	$1.18^*_{(.56)}$	.58	$1.11^*_{(.53)}$	.64	$1.17^*_{(.56)}$	.59

*Cases in which the observed bias is significant at the 5% level.

Sinha (2004) studied outliers in the covariate x. He considered Poisson and binary GLMMs (8) with $n = 100$, $J_i = 4$, and $\beta_0 = 0$. Outliers were created in each data set by moving four randomly chosen points (i.e., 1 % of the data points) from the bulk of the data toward the x direction. More specifically, to create outliers, Sinha replaced the corresponding x values with $x + 3$ and the y values with 0 in the case of the binary GLMM with $\beta_1 = 1.5$ and $\lambda = 1$. For the Poisson GLMM, these outliers were created by replacing the corresponding x values by $x - 3$ with $\beta_1 = 1$ and $\lambda = .25$. Results for Sinha's method are from his article. To account for these outliers among the x values, Sinha proposed weighted score equations. Our current robust methods do not consider the weights w_i in (5) that depend on x_i, so that they are not fully robust against such outliers. We can, however, accommodate Sinha's method by using estimating equations

$$\sum_{i=1}^{n} v_i(y_i) w_i x_i = 0, \tag{9}$$

where w_i is Sinha's (2004) weight in his (8). From Table 2, we can see that this adjustment of score equations gives robust results against outliers among the x values; Sinha's method corresponds to our method using fixed τ and w. Using the REML procedure, we again have robust estimation without losing efficiency in the absence of outliers. In the binary GLMM, the outliers can cause upward biases with robust methods for estimating λ.

7. EXAMPLES

7.1 Scottish Data on Lip Cancer Rates

Clayton and Kaldor (1997) analyzed observed (y_i) and expected numbers (n_i) of lip cancer cases in the 56 administrative areas of Scotland with a view toward producing a map that displays regional variations in cancer incidence and yet avoids the presentation of unstable rates for the smaller areas. Data were available on the percentages of the work force in each area employed in agriculture, fishing, and forestry (x_i). This covariate exhibits spatial aggregation paralleling that for lip cancer itself. For analysis, Breslow and Clayton (1993) considered the following Poisson HGLM with log link:

$$\eta_i = \log \mu_i = \log n_i + \beta_0 + \beta_1 x_i/10 + v_i,$$

where $\mathbf{v} = (v_1, \ldots, v_n)^t \sim \exp\{-\sum_{i\sim j}(v_i - v_j)^2/(2\lambda)\}$ and $i \sim j$ denotes adjacent counties. Because of the need for high-dimensional integration, Sinha's (2004) method cannot be used. Yau and Kuk's PQL method is possible but may not be reliable because of biases. The results of fitting our method are given in Table 3. Figure 8 shows the normal probability plot for the standardized residuals based on the ML method. The two largest

Table 3. Summaries of analysis for the Scottish lip cancer data

	Classical ML method: Full data Estimate	Classical ML method: Full data SE	Classical ML method: Without two outliers Estimate	Classical ML method: Without two outliers SE	Robust method using fixed τ Estimate	Robust method using fixed τ SE	Robust method using REML Estimate	Robust method using REML SE
β_0	−.20	.12	−.32	.13	−.30	.13	−.31	.13
β_1	.35	.13	.46	.13	.44	.12	.44	.12
λ	.55		.55		.56		.56	
Elapsed time (seconds)	2.4		2.3		3.6		10.5	

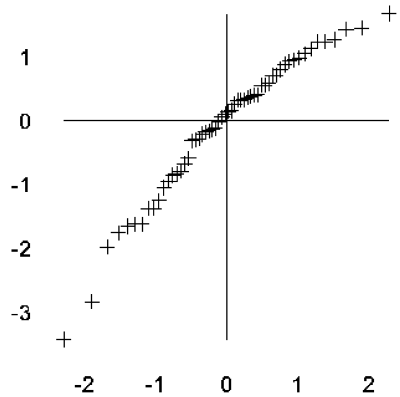

Figure 8. Normal probability plot for the standardized residuals using full data.

negative outliers are caused by two zero responses. Our robust results are similar to those from the ML method without the two outliers.

7.2 Damaged Carrots Data

Using a data set on damaged carrots, McCullagh and Nelder (1989) illustrated model-checking techniques for isolated departures from the model caused by the presence of an outlier among the y values. The data came from a soil experiment and gave the proportion of carrots showing insect damage in a trial with three blocks and eight dosage levels of insecticide. Cantoni and Ronchetti (2001) considered a binomial GLM,

$$\text{logit}(p_i) = \beta_0 + \beta_1 \log(\text{dose}) + \beta_2 \text{block}_2 + \beta_3 \text{block}_1,$$

where p_i is the probability that the tree is damaged by insect in the ith trial and block_j for $j = 1, 2$ are indicator variables taking value 1 if measures are taken in block_j and 0 otherwise. We take the effect of the block 3 to be null. Table 4 gives the results for our robust methods, the ML method from the GLM, and Cantoni and Ronchetti's (2001) robust method.

Various model-checking techniques show that there is a large outlier—observation 14 (dose level 6 and block 2). However, this observation does not appear with high leverage, because the corresponding GLM leverage value is small. Table 4 shows the effect of observation 14. It seems to decrease the value of β_0 and increases the values of β_1, β_2, and β_3 in nonrobust estimation.

From Table 5, the classical likelihood ratio test of the GLM shows that all of the variables, added sequentially, are highly significant on the basis of their deviance value. Cantoni and Ronchetti (2001) noted that the corresponding deviance test for the covariate block_1, based on their quasi-likelihood, is not significant at size .05. The results of the deviance tests, based on the likelihood of our robust model, agree with the findings of Cantoni and Ronchetti (2001).

Table 4. Estimate (SE) of $\boldsymbol{\beta}$ for the damaged carrots data

Effects	Classical ML method	Cantoni and Ronchetti's method	Robust method using fixed τ	Robust method using REML
Intercept	$1.480_{(.66)}$	$1.939_{(.70)}$	$2.042_{(.73)}$	$2.047_{(.73)}$
log (dose)	$-1.817_{(.34)}$	$-2.049_{(.37)}$	$-2.120_{(.38)}$	$-2.123_{(.38)}$
block_2	$.834_{(.23)}$	$.685_{(.24)}$	$.661_{(.24)}$	$.659_{(.24)}$
block_1	$.542_{(.23)}$	$.450_{(.24)}$	$.487_{(.24)}$	$.462_{(.25)}$
Elapsed time (seconds)	.5	1.4	1.1	3.7

7.3 Epilepsy Seizure Count Data

Thall and Vail (1990) presented longitudinal data from a clinical trial of 59 epileptics, randomized to a new drug or a placebo ($Trt = 1$ or $Trt = 0$). Baseline data available at the start of the trial included the logarithm of the average number of epileptic seizures recorded in the 8-week period preceding the trial (*Base*), the logarithm of age (*Age*), and fourth visit or not ($V_4 = 1$ or $V_4 = 0$). A multivariate response variable was the seizure count during each the 2-week period before each of the four clinic visits.

Yau and Kuk (2002) considered two Poisson GLMMs for $i = 1, \ldots, 59$ and $j = 1, \ldots, 4$, a model with one random component, $\eta_{ij} = \mathbf{x}_{ij}^t \boldsymbol{\beta} + v_i$, and a model with two random components, $\eta_{ij} = \mathbf{x}_{ij}^t \boldsymbol{\beta} + v_i + v_{ij}$, where $v_i \sim \text{N}(0, \lambda_1)$ and $v_{ij} \sim \text{N}(0, \lambda_2)$.

The results are given in Table 6. With the one–random-component model, Yau and Kuk observed different results for V_4 from the robust and nonrobust methods. Because of the nonignorable biases of their method, it is useful to investigate this problem using our method, which does not have such a bias (Sec. 6.2). Table 6 confirms their findings.

Figure 9(a) shows the normal probability plot for the standardized residuals. The largest outlier corresponds to the third observation of patient 227. Yau and Kuk (2002) observed that patient 225 has a large patient effect. The scatterplot of robust estimates using a fixed tuning parameter for the patient effects versus nonrobust estimates is shown in Figure 10(a). With nonrobust estimation, patient 225 has the largest patient effect, whereas with robust estimation, this patient has the second-largest patient effect. The second largest patient effect in robust estimation (for patient 207) was the fifth-largest effect in nonrobust estimation. As shown in Figure 10, the robust method clearly shrinks the extreme patient effects more than those for less-extreme effects. With the two–random-component model, Yau and Kuk (2002) found that the robust and nonrobust methods gave similar results. Figure 9(b) shows no apparent outlier.

With the two–random-component model, Figure 10(b) shows that the rank changes are not dramatic, even though the robust estimation still shrinks extreme effects more. The robust method using REML estimation estimates the value of the tuning parameter τ to be near 0, implying that there is no potential outlier among the y values. We also tried the robust methods with Sinha's (2004) weights and found that they gave identical results to those without weights, again showing that there are no outliers among the x values.

Table 5. Deviance difference (p value) for the damaged carrots data

Effects	Classical ML method	Cantoni and Ronchetti's method	Robust method using fixed τ	Robust method using REML
NULL	0	0	0	0
log (dose)	$28.61_{(0)}$	$20.54_{(0)}$	$21.36_{(0)}$	$21.40_{(0)}$
block_2	$9.14_{(.003)}$	$4.73_{(.030)}$	$4.59_{(.032)}$	$4.56_{(.033)}$
block_1	$5.61_{(.018)}$	$2.47_{(.116)}$	$2.86_{(.091)}$	$2.70_{(.096)}$

Table 6. Summaries of analysis for the epileptics data

	Classical ML method		Yau and Kuk's method		Robust method using fixed τ		Robust method using REML	
	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE
One–random-component model								
Constant	−1.32	1.24	−1.19	1.32	−1.29	1.30	−1.27	1.35
Base	.88	.14	.84	.15	.86	.15	.85	.15
Trt	−.94	.42	−.93	.44	−.94	.43	−.94	.43
Base.Trt	.34	.21	.34	.23	.34	.22	.34	.22
Age	.48	.36	.46	.39	.47	.38	.47	.38
V_4	−.16	.05	−.11	.06	−.11	.06	−.10	.06
λ_1	.29		.29		.30		.31	
Elapsed time (seconds)	3.6		7.3		7.7		25.0	
Two–random-component model								
Constant	−1.39	1.22	−1.25	1.31	−1.38	1.23	−1.39	1.22
Base	.88	.14	.85	.15	.88	.14	.88	.14
Trt	−.95	.41	−.93	.44	−.95	.42	−.95	.41
Base.Trt	.35	.21	.34	.22	.35	.22	.35	.21
Age	.48	.36	.47	.38	.48	.37	.48	.36
V_4	−.10	.09	−.10	.09	−.10	.09	−.10	.09
λ_1	.24		.25		.25		.24	
λ_2	.13		.14		.14		.14	
Elapsed time (seconds)	1.2		19.7		21.3		49.8	

8. CONCLUDING REMARKS

We introduce robust statistical models for GLM classes, that give robust estimation with bounded influence for the parameters, as well as random effects in GLMMs. Using REML estimation for the tuning parameter allows the robust estimation method to maintain high efficiency for noncontaminated data. The robust method downweights the outlying observations among the y values. By allowing weights in the estimating equations, we also can downweight outliers among the x values. Our method can be applied to a wide range of models, including models with more than one random component, models with nonnormal random effects, and random-effects models with temporal and spatial correlations. The method converged well without divergence throughout all of our numerical studies. Recently, Copt and Victoria-Feser (2006) proposed a high-breakdown robust estimator for linear mixed models using Tukey's biweight function. It would be interesting to develop robust modeling with a high breakdown in nonnormal GLM classes.

APPENDIX A: PROOF OF THEOREM 1

In the Poisson GLM class, $\tilde{b_i}$ is the solution of

$$\begin{aligned}\frac{\partial h_i}{\partial b_i} &= \phi_i\left[\frac{y_i}{\phi_i} - \frac{y_i\mu_i}{1+\phi_i\mu_i} + \frac{\log(1+\phi_i\mu_i)}{\phi_i^2}\right.\\ &\quad - \frac{\mu_i}{\phi_i(1+\phi_i\mu_i)} - \frac{1+1/\tau}{\phi_i} + \frac{1}{\phi_i^2}\\ &\quad \left. - \frac{1}{\phi_i^2}dg\left(y_i + \frac{1}{\phi_i}\right) + \frac{1}{\phi_i^2}dg\left(\frac{1}{\phi_i}\right)\right],\end{aligned}$$

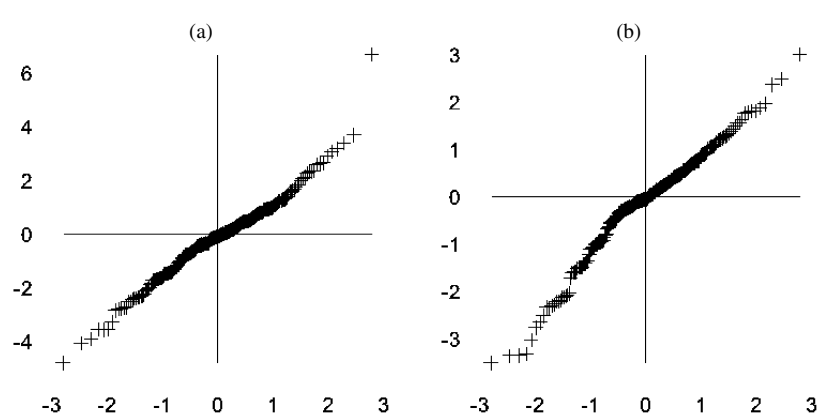

Figure 9. Normal probability plot for the standardized residuals in the one–random-component model and (a) two–random-component model (b) based on the classical ML method.

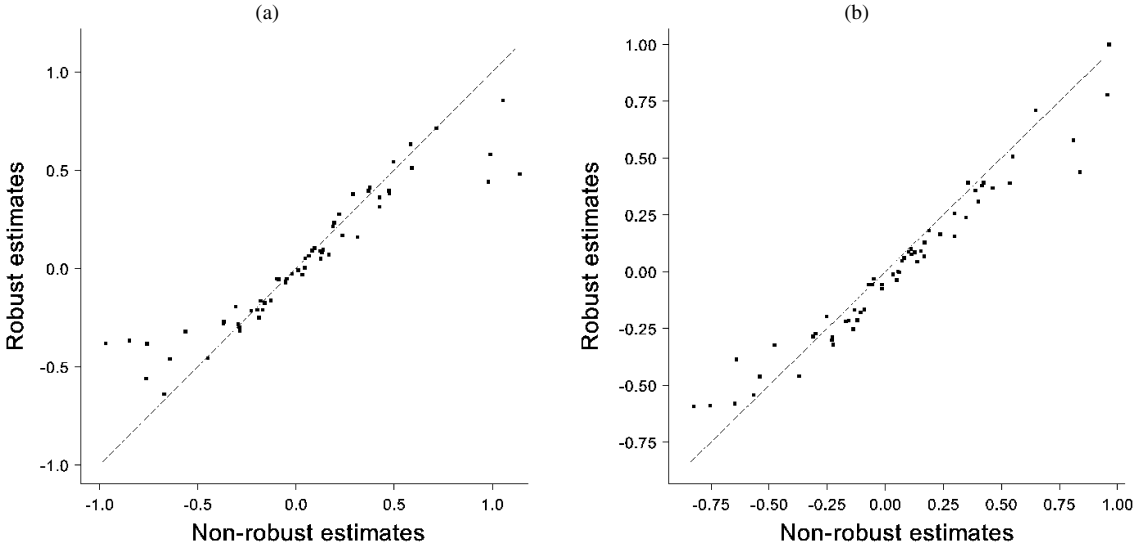

Figure 10. Scatterplots of robust estimates versus GLMM estimates for patient effect in the one–random-component model (a) and two–random-component model (b).

where $h_i = y_i \log(\phi_i \mu_i) - y_i \log(1 + \phi_i \mu_i) - \log(1 + \phi_i \mu_i)/\phi_i - (1 + 1/\tau)\log(\phi_i) - 1/\phi_i + \log\Gamma(y_i + 1/\phi_i) - \log\Gamma(y_i + 1) - \log\Gamma(1/\phi_i) + \log(\tau) - \log\Gamma(1 + 1/\tau)$ and $dg(\cdot)$ is the di-gamma function.

Let $c_i = \tilde{\phi}_i / y_i$. We use the results that $dg(x) = O(\log(x))$ and $tg(1/x) = O(x^2)$ for large x (Abramowitz and Stegun 1972). For large y_i, we have

$$\begin{aligned}\left.\frac{\partial h_i}{\partial b_i}\right|_{\phi_i=\tilde{\phi}_i} &= y_i - \frac{y_i \mu_i \tilde{\phi}_i}{1+\tilde{\phi}_i} - \left(1 + \frac{1}{\tau}\right) + O\left(\frac{1}{y_i}\right) \\ &= \frac{y_i}{1 + c_i y_i \mu_i} - \left(1 + \frac{1}{\tau}\right) + O\left(\frac{1}{y_i}\right) \\ &= 0.\end{aligned}$$

Suppose that y_i goes to infinity. Then $c_i = 1/\{\mu_i(1 + 1/\tau)\}$, so that $\tilde{\phi}_i = \exp(\tilde{b}_i) = O(y_i)$.

Note that the term $\nu_i(y_i)$ in (4) is

$$\nu_i(y_i) = \left(\left.\frac{\partial h_i}{\partial \mu_i}\right|_{b_i=\tilde{b}_i} - \frac{1}{2}\frac{\partial}{\partial \mu_i}\left[\log\{D(h_i, b_i)\}\right]\Big|_{b_i=\tilde{b}_i}\right)\frac{\partial \mu_i}{\partial \eta_i},$$

where

$$\begin{aligned}D(h_i, b_i) &= \frac{\phi_i y_i \mu_i}{(1+\phi_i\mu_i)^2} + \frac{1}{\phi_i}\log(1+\phi_i\mu_i) - \frac{\mu_i}{1+\phi_i\mu_i} \\ &\quad - \frac{\phi_i \mu_i^2}{(1+\phi_i\mu_i)^2} + \frac{1}{\phi_i} - \frac{1}{\phi_i} dg\left(y_i + \frac{1}{\phi_i}\right) \\ &\quad - \frac{1}{\phi_i} tg\left(y_i + \frac{1}{\phi_i}\right) + \frac{1}{\phi_i} dg\left(\frac{1}{\phi_i}\right) + \frac{1}{\phi_i} tg\left(\frac{1}{\phi_i}\right)\end{aligned}$$

and $tg(\cdot)$ is the tri-gamma function. Because $\tilde{\phi}_i = O(y_i)$, we can show that $D(h_i, b_i)|_{b_i=\tilde{b}_i} = O(y_i)$, $\partial D(h_i, b_i)/\partial \mu_i|_{b_i=\tilde{b}_i} = O(1)$, and $\partial D(h_i, b_i)/\partial b_i|_{b_i=\tilde{b}_i} = O(y_i^2)$.

Because $\{\partial^2 h_i/(\partial \mu_i\, \partial b_i)\}|_{b_i=\tilde{b}_i} + \{\partial^2 h_i/(\partial b_i)^2\}|_{b_i=\tilde{b}_i}\partial \tilde{b}_i/\partial \mu_i = 0$, we have

$$\partial \tilde{b}_i/\partial \mu_i = O(1/y_i).$$

Thus $\nu_i(y_i) = O(1)$, because

$$\left.\frac{\partial h_i}{\partial \mu_i}\right|_{b_i=\tilde{b}_i} = \frac{y_i - \mu_i}{\mu_i(1+\tilde{\phi}_i\mu_i)} = O(1)$$

and

$$\begin{aligned}&\frac{\partial}{\partial \mu_i}\left[\log\{D(h_i, b_i)\}\right]\Big|_{b_i=\tilde{b}_i} \\ &\quad = \frac{\{\partial D(h_i, b_i)/\partial \mu_i\}|_{b_i=\tilde{b}_i} + \{\partial D(h_i, b_i)/\partial b_i\}|_{b_i=\tilde{b}_i}\partial \tilde{b}_i/\partial \mu_i}{D(h_i, b_i)|_{b_i=\tilde{b}_i}} \\ &\quad = \frac{O(1) + O(y_i^2)O(1/y_i)}{O(y_i)} = O(1).\end{aligned}$$

APPENDIX B: PROOF OF THEOREM 2

In the binomial GLM class, $\tilde{b}_i$ is the solution of

$$\begin{aligned}\frac{\partial h_i}{\partial b_i} &= \phi_i\left\{\sum_{k=0}^{y_i-1}\frac{k}{p_i + k\phi_i} + \sum_{k=0}^{m_i-y_i-1}\frac{k}{1-p_i+k\phi_i}\right. \\ &\quad \left. - \sum_{k=0}^{m_i-1}\frac{k}{1+k\phi_i} - \frac{1+1/\tau}{\phi_i} + \frac{1}{\phi_i^2}\right\} \\ &= 0,\end{aligned}$$

where $h_i = \sum_{k=0}^{y_i-1}\log(p_i + k\phi_i) + \sum_{k=0}^{m_i-y_i-1}\log(1 - p_i + k\phi_i) - \sum_{k=0}^{m_i-1}\log(1 + k\phi_i) + \log\binom{m_i}{y_i} - (1 + 1/\tau)\log(\phi_i) - 1/\phi_i + \log(\tau)$. Suppose that y_i approaches m_i with m_i going to infinity. Let $c_i = \tilde{\phi}_i/\log(m_i)$. Then we have that

$$\begin{aligned}\left.\frac{\partial h_i}{\partial b_i}\right|_{b_i=\tilde{b}_i} &= \sum_{k=0}^{m_i-1}\frac{k\tilde{\phi}_i}{p_i + k\tilde{\phi}_i} - \sum_{k=0}^{m_i-1}\frac{k\tilde{\phi}_i}{1+k\tilde{\phi}_i} \\ &\quad - \left(1 + \frac{1}{\tau}\right) + O\left(\frac{1}{\log(m_i)}\right) \\ &= \sum_{k=0}^{m_i-1} -\frac{p_i}{p_i + kc_i\log(m_i)} + \sum_{k=0}^{m_i-1}\frac{1}{1+kc_i\log(m_i)} \\ &\quad - \left(1 + \frac{1}{\tau}\right) + O\left(\frac{1}{\log(m_i)}\right)\end{aligned}$$

$$= -\frac{p_i}{c_i}O(1) + \frac{1}{c_i}O(1) - \left(1+\frac{1}{\tau}\right) + O\left(\frac{1}{\log(m_i)}\right)$$
$$= 0,$$

using the fact (Abramowitz and Stegun 1972) that

$$\sum_{k=0}^{m_i-1} \frac{1}{a+kb} = \frac{1}{b}O(\log(m_i)).$$

Thus $c_i = O(1)$, so that $\tilde{\phi}_i = O(\log(m_i))$. Similarly, we can show that $\tilde{\phi}_i = O(\log(m_i))$ as y_i approaches 0.

Here the term $\nu_i(y_i)$ is

$$\nu_i(y_i) = \left(\frac{\partial h_i}{\partial p_i}\bigg|_{b_i=\tilde{b}_i} - \frac{1}{2}\frac{\partial}{\partial p_i}\big[\log\{D(h_i,b_i)\}\big]\bigg|_{b_i=\tilde{b}_i}\right)\frac{\partial p_i}{\partial \eta_i},$$

where

$$D(h_i,b_i) = -\sum_{k=0}^{y_i-1}\frac{kp_i\phi_i}{(p_i+k\phi_i)^2} - \sum_{k=0}^{m_i-y_i-1}\frac{k(1-p_i)\phi_i}{(1-p_i+k\phi_i)^2} + \sum_{k=0}^{m_i-1}\frac{k\phi_i}{(1+k\phi_i)^2} + \frac{1}{\phi_i}.$$

If y_i approaches 0 or m_i, with m_i going to infinity, then we have $D(h_i,b_i)|_{b_i=,\tilde{b}_i} = O(1)$, $[\partial D(h_i,b_i)/\partial p_i]|_{b_i=\tilde{b}_i} = O(1)$, $[\partial D(h_i, b_i)/\partial b_i]|_{b_i=\tilde{b}_i} = O(1)$, and $\partial\tilde{b}_i/\partial p_i = O(1)$.

Suppose that m_i goes to infinity. If $y_i = m_i$, then

$$\frac{\partial h_i}{\partial p_i}\bigg|_{\phi_i=\tilde{\phi}_i} = \sum_{k=0}^{m_i-1}\frac{1}{p_i+kc_i\log(m_i)} = O(1),$$

and if $y_i = 0$, then

$$\frac{\partial h_i}{\partial p_i}\bigg|_{\phi_i=\tilde{\phi}_i} = -\sum_{k=0}^{m_i-1}\frac{1}{1-p_i+kc_i\log(m_i)} = O(1).$$

Thus $\nu_i(y_i) = O(1)$, because

$$\frac{\partial}{\partial p_i}\big[\log\{D(h_i,b_i)\}\big]\bigg|_{b_i=\tilde{b}_i} = \frac{\{\partial D(h_i,b_i)/\partial p_i\}|_{b_i=\tilde{b}_i} + \{\partial D(h_i,b_i)/\partial b_i\}|_{b_i=\tilde{b}_i}\partial\tilde{b}_i/\partial p_i}{D(h_i,b_i)|_{b_i=\tilde{b}_i}} = O(1).$$

This completes the proof.

APPENDIX C: PROOFS OF THEOREMS 3 AND 4

In the Poisson GLMM class, $\tilde{v}_i$ is the solution of

$$\frac{\partial h_i}{\partial v_i} = \sum_j^{J_i}\frac{y_{ij}-\mu_{ij}}{1+\phi_{ij}\mu_{ij}} - \frac{v_i}{\lambda} = 0,$$

where $h_i = \sum_{j=1}^{J_i}\{y_{ij}\log(\phi_{ij}\mu_{ij}) - y_{ij}\log(1+\phi_{ij}\mu_{ij}) - \log(1+\phi_{ij}\mu_{ij})/\phi_{ij} - (1+1/\tau)\log(\phi_{ij}) - 1/\phi_{ij} + \log\Gamma(y_{ij}+1/\phi_{ij}) - \log\Gamma(y_{ij}+1) - \log\Gamma(1/\phi_{ij}) + \log(\tau) - \log\Gamma(1+1/\tau)\} - v_i^2/(2\lambda) - \frac{1}{2}\log(2\pi\lambda)$.

Similar to Appendix A, we can show that $\tilde{\phi}_{ij} = O(y_{ij})$ and $\tilde{v}_i = O(1)$ with y_{i+} going to infinity because

$$\tilde{v}_i = \lambda\sum_j^{J_i}\frac{y_{ij}-\tilde{\mu}_{ij}}{1+\tilde{\phi}_{ij}\tilde{\mu}_{ij}} = O(1),$$

where $\tilde{\mu}_{ij} = g^{-1}(\mathbf{x}_{ij}^t\boldsymbol{\beta} + \tilde{v}_i)$.

In the binomial GLMM class, $\tilde{v}_i$ is the solution of

$$\frac{\partial h_i}{\partial v_i} = \sum_j^{J_i}\sum_{k=0}^{y_{ij}-1}\frac{1}{p_{ij}+k\phi_{ij}} - \sum_j^{J_i}\sum_{k=0}^{m_{ij}-y_{ij}-1}\frac{1}{1-p_{ij}+k\phi_{ij}} - \frac{v_i}{\lambda} = 0,$$

where $h_i = \sum_{j=1}^{J_i}[\sum_{k=0}^{y_{ij}-1}\log(p_{ij}+k\phi_{ij}) + \sum_{k=0}^{m_{ij}-y_{ij}-1}\log(1-p_{ij}+k\phi_{ij}) - \sum_{k=0}^{m_{ij}-1}\log(1+k\phi_{ij}) + \log\binom{m_{ij}}{y_{ij}} - (1+1/\tau)\log(\phi_{ij}) - 1/\phi_{ij} + \log(\tau)] - v_i^2/(2\lambda) - 1/2\log(2\pi\lambda)$.

Similar to Appendix B, we can show that $\tilde{\phi}_{ij} = O(\log(m_{ij}))$ as m_{ij} goes to infinity. Suppose that m_{i+} goes to infinity. If $y_{ij} = m_{ij}$, then

$$\tilde{v}_i = \sum_j^{J_i}\sum_{k=0}^{m_{ij}-1}\frac{\lambda}{\tilde{p}_{ij}+k\tilde{\phi}_{ij}} = \sum_j^{J_i}\frac{\lambda}{\tilde{\phi}_{ij}}O(\log(m_{ij})) = O(1),$$

and if $y_{ij} = 0$

$$\tilde{v}_i = -\sum_j^{J_i}\sum_{k=0}^{m_{ij}-1}\frac{\lambda}{1-\tilde{p}_{ij}+k\tilde{\phi}_{ij}} = \sum_j^{J_i}\frac{\lambda}{\tilde{\phi}_{ij}}O(\log(m_{ij})) = O(1),$$

where $\tilde{p}_{ij} = g^{-1}(\mathbf{x}_{ij}^t\boldsymbol{\beta} + \tilde{v}_i)$. Similar to Appendixes A and B, we can show that $\partial p_{\mathbf{v},\mathbf{b}}(h_i)/\partial\beta_k = O(1)$ for all k and $\partial p_{\boldsymbol{\beta},\mathbf{v},\mathbf{b}}(h_i)/\partial\lambda = O(1)$.

APPENDIX D: PROOFS OF THEOREMS 5–7

Proof of Theorem 5

In the Poisson GLM class of Section 2.1, let $L_i(\boldsymbol{\theta},\tau) = \int f(y_i|\phi_i;\boldsymbol{\theta},\tau)f(\phi_i;\tau)\,d\phi_i = f(y_i;\boldsymbol{\theta},\tau)$, where $f(y_i|\phi_i;\boldsymbol{\theta},\tau)$ is the density function of the negative-binomial distribution (5) and $f(\phi_i;\tau)$ is that for ϕ_i. Let $E_\tau\{g(y_i;\boldsymbol{\theta},\tau_0)\} = \int g(y_i;\boldsymbol{\theta},\tau_0)f(y_i;\boldsymbol{\theta},\tau)\,dy_i$. Under appropriate regularity conditions, we have

$$\begin{aligned}
&E_\tau\left\{\frac{\partial\log L_i(\boldsymbol{\theta},\tau_0)}{\partial\mu_i}\right\}\\
&= E_\tau\left\{\frac{\partial f(y_i;\boldsymbol{\theta},\tau_0)/\partial\mu_i}{f(y_i;\boldsymbol{\theta},\tau_0)}\right\}\\
&= E_\tau\left[\left\{\int\frac{y_i-\mu_i}{\mu_i(1+\phi_i\mu_i)}\frac{f(y_i|\phi_i;\boldsymbol{\theta},\tau_0)f(\phi_i;\boldsymbol{\theta},\tau_0)}{f(y_i;\boldsymbol{\theta},\tau_0)}\,d\phi_i\right\}\right]\\
&= E_\tau\left[\left\{\int\frac{y_i-\mu_i}{\mu_i(1+\phi_i\mu_i)}f(\phi_i|y_i;\boldsymbol{\theta},\tau_0)\,d\phi_i\right\}\right]\\
&= E_\tau\left[E_{\tau_0}\left\{\frac{y_i-\mu_i}{\mu_i(1+\phi_i\mu_i)}\bigg|y_i\right\}\right]\\
&= E_\tau\left\{\frac{y_i-\mu_i}{\mu_i(1+\phi_i\mu_i)}\right\}\\
&= E_\tau\left[E_\tau\left\{\frac{y_i-\mu_i}{\mu_i(1+\phi_i\mu_i)}\bigg|\phi_i\right\}\right] = 0,
\end{aligned}$$

because $E_\tau(y_i|\phi_i) = \mu_i$ for all $\tau \geq 0$. Thus

$$E_\tau\left\{\frac{\partial l(\boldsymbol{\theta},\tau_0)}{\partial\boldsymbol{\beta}}\right\} = \sum_{i=1}^n E_\tau\left\{\frac{\partial\log L_i(\boldsymbol{\theta},\tau_0)}{\partial\mu_i}\frac{\partial\mu_i}{\partial\boldsymbol{\beta}}\right\} = \mathbf{0},$$

so that the robust estimators for $\boldsymbol{\beta}$ are consistent for all $\tau \geq 0$.

In the Poisson GLMM class of Section 3.1, we can similarly show that the robust estimators for $\boldsymbol{\beta}$ are consistent for all $\tau \geq 0$. Further-

more, because $E(v_i^2) = \lambda$ and

$$\frac{\partial l(\boldsymbol{\theta}, \tau_0)}{\partial \lambda} = \sum_{i=1}^{n} \int \left\{ \Pi_j f(y_{ij}|v_i, \phi_{ij}; \boldsymbol{\theta}, \tau_0) f(\phi_{ij}; \tau_0) \right\} \times \frac{v_i^2 - \lambda}{2\lambda^2} \frac{f(v_i)}{f(y_i; \boldsymbol{\theta}, \tau_0)} \Pi_j \, d\phi_{ij} \, dv_i,$$

we can show that $E_\tau(\partial l(\boldsymbol{\theta}, \tau_0)/\partial \lambda) = 0$ for all $\tau \geq 0$. Thus the robust estimator for λ is also consistent for all $\tau \geq 0$.

Proof of Theorem 6

Let $f(y_i|\phi_i; \boldsymbol{\theta}, \tau)$ be the density function of the beta-binomial distribution (6) in the binomial GLM class of Section 2.2. Then we have

$$\frac{\partial \log f(y_i|\phi_i; \boldsymbol{\theta}, \tau_0)}{\partial p_i} = \sum_{k=0}^{y_i - 1} \frac{1}{p_i + k\phi_i} - \sum_{k=0}^{m_i - y_i - 1} \frac{1}{1 - p_i + k\phi_i}.$$

When $0 < p_i < 1$ and m_i goes to infinity, both y_i and $m_i - y_i$ go to infinity. Let $q_i = 1 - p_i$. Then we have

$$\begin{aligned}
&\lim_{m_i \to \infty} \frac{\partial \log f(y_i|\phi_i; \boldsymbol{\theta}, \tau_0)}{\partial p_i} \\
&= \frac{1}{\phi_i} \lim_{m_i \to \infty} \left[\left\{ \log\left(y_i - 1 + \frac{p_i}{\phi_i} \right) - \log\left(\frac{p_i}{\phi_i} \right) \right\} \right. \\
&\quad \left. - \left\{ \log\left(m_i - y_i - 1 + \frac{q_i}{\phi_i} \right) - \log\left(\frac{q_i}{\phi_i} \right) \right\} \right] \\
&= \frac{1}{\phi_i} \lim_{m_i \to \infty} \left[\left\{ \log\left(\frac{y_i - 1 + p_i/\phi_i}{m_i} \right) - \log(p_i) \right\} \right. \\
&\quad \left. - \left\{ \log\left(\frac{m_i - y_i - 1 + q_i/\phi_i}{m_i} \right) - \log(q_i) \right\} \right] \\
&= \frac{1}{\phi_i} \lim_{m_i \to \infty} \left\{ \log\left(\frac{y_i}{m_i} \right) - \log(p_i) - \log\left(1 - \frac{y_i}{m_i} \right) + \log(q_i) \right\} \\
&= 0 \quad \text{in probability,}
\end{aligned}$$

using the fact (Abramowitz and Stegun 1972) that

$$\lim_{m_i \to \infty} \sum_{k=0}^{y_i - 1} \frac{1}{p_i + k\phi_i} = \lim_{m_i \to \infty} \frac{1}{\phi_i} \left\{ \log\left(y_i - 1 + \frac{p_i}{\phi_i} \right) - \log\left(\frac{p_i}{\phi_i} \right) \right\} + c$$

and

$$\lim_{m_i \to \infty} \sum_{k=0}^{m_i - y_i - 1} \frac{1}{1 - p_i + k\phi_i} = \lim_{m_i \to \infty} \frac{1}{\phi_i} \left\{ \log\left(m_i - y_i - 1 + \frac{q_i}{\phi_i} \right) - \log\left(\frac{q_i}{\phi_i} \right) \right\} + c.$$

Thus, for all $\tau \geq 0$,

$$\begin{aligned}
&E_\tau \left\{ \lim_{m_i \to \infty} \frac{\partial l(\boldsymbol{\theta}, \tau_0)}{\partial \boldsymbol{\beta}} \right\} \\
&= \sum_{i=1}^{n} E_\tau \left\{ \lim_{m_i \to \infty} \frac{\partial f(y_i; \boldsymbol{\theta}, \tau_0)/\partial p_i}{f(y_i; \boldsymbol{\theta}, \tau_0)} \frac{\partial p_i}{\partial \boldsymbol{\beta}} \right\} \\
&= \sum_{i=1}^{n} E_\tau \left\{ \int \lim_{m_i \to \infty} \frac{\partial \log f(y_i|\phi_i; \boldsymbol{\theta}, \tau_0)}{\partial p_i} \right. \\
&\quad \left. \times f(\phi_i|y_i; \boldsymbol{\theta}, \tau_0) \frac{\partial p_i}{\partial \boldsymbol{\beta}} \, d\phi_i \right\} = \mathbf{0}.
\end{aligned}$$

Similary, in the binomial GLMM class of Section 3.2, we can show that for all $\tau \geq 0$,

$$E_\tau \left(\lim_{m_{ij} \to \infty} \frac{\partial \log f(y_{ij}|\phi_{ij}; \boldsymbol{\theta}, \tau_0)}{\partial p_{ij}} \Big| \phi_{ij}, v_i \right) = 0,$$

if $0 < p_{ij} < 1$. Similar, to the proof of Theorem 5, we also can show that the robust estimator for λ is consistent for all $\tau \geq 0$.

Proof of Theorem 7

Self and Liang (1987) gave a consistency result for the MLE (and thus REML estimator) $\hat{\tau}$ when the true parameter value $\tau = \tau_0$ is on the boundary of the parameter space. It can be shown that when $\tau_0 = 0$, the log-likelihoods $l(\boldsymbol{\theta}, \tau)$ of the Poisson and binomial GLM classes in Sections 2 and 3 satisfy the following conditions:

a. The almost sure existence of the first three derivatives of $l(\boldsymbol{\theta}, \tau)$ with respect to τ on the interception of neighbors of τ_0 and the parameter space Ω.

b. On the interception of neighbors of τ_0 and Ω, n^{-1} times the absolute value of the third derivative of $l(\boldsymbol{\theta}, \tau)$ is bounded by a function of $y_1, \ldots, y_n$, whose expectation exists.

c. $E(-\frac{1}{n}\partial^2 l(\boldsymbol{\theta}, \tau)/\partial \tau^2)$ is positive definite on neighbors of τ_0. Thus, from theorem 1 of Self and Liang (1987), $\hat{\tau}$ converges to $\tau_0 = 0$ in probability.

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